

Problem sheet 3: Non-local Games

Problem 1 (HW). Shared randomness. Suppose that there exists a classical strategy with shared randomness, winning a non-local game $\mathcal{G}$ with probability p_{win} . Prove that there is also a deterministic strategy winning $\mathcal{G}$ with probability at least p_{win} . *For simplicity you can assume that the shared randomness is just a discrete random variable with finite set of outcomes.*

Problem 2 (HW). Shared separables state. Suppose that there is a quantum strategy using a separable quantum state ρ_{AB} , winning a non-local game $\mathcal{G}$ with probability p_{win} . Prove that there also exists a randomized (and hence deterministic) strategy winning $\mathcal{G}$ with probability at least p_{win} .

Problem 3 (HW). Correlations beyond quantum theory. Find a non-signalling strategy that always wins the CHSH game. *There is no physical evidence for a theory supporting such correlations. So, do not worry about how to realize such a strategy, just describe the strategy as $p = (p(a, b|x, y)) \in [0, 1]^{O_A \times O_B \times I_A \times I_B}$.*

Problem 4. The GHS game. Let $\mathcal{H}_{ABC} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ be a Hilbert space of three qubits shared between three parties. The *Greenberger-Horne-Zeilinger (GHZ)* state is

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle).$$

(a) Show that the reduced state on A is given by

$$\text{tr}_{BC}[|\text{GHZ}\rangle\langle\text{GHZ}|] = \frac{1}{2}\mathbb{1}_A.$$

So, the reduced state on A is the maximally mixed state.

(b) Define the operators

$$\begin{aligned} M_0 &= X \otimes X \otimes X, \\ M_1 &= X \otimes Y \otimes Y, \\ M_2 &= Y \otimes X \otimes Y, \\ M_3 &= Y \otimes Y \otimes X, \end{aligned}$$

where X and Y are Pauli matrices. Show that $|\text{GHZ}\rangle$ is an eigenstate of each of these operators, determine the corresponding eigenvalues λ_i .

The GHZ game is *three-player* Bell game in which referee gives a question $x, y, z \in \{0, 1\}$ to each of Alice, Bob, and Charlie, respectively. The referee either gives 0 to all three players, or

gives 0 to one of them and 1 to the other two. The players output $a, b, c \in \{0, 1\}$, respectively. The winning condition is

$$x \wedge y \wedge z = a \oplus b \oplus c$$

($\wedge$ is the logical and, $\oplus$ is addition modulo 2).

- (c) Suppose that Alice, Bob, and Charlie use classical deterministic strategy. By considering the sum

$$\sum_{x,y,z} (f(x) + g(y) + h(z)) \pmod{2},$$

show that the maximum winning probability is $\omega(\text{GHZ}) = 3/4$.

- (d) Now suppose that Alice, Bob, and Charlie share 3-party state $|\text{GHZ}\rangle$. Each of them adopts the following strategy: If 0 is received, measure X . If 1 is received, measure Y . The resulting measurement gives outcome ± 1 ; this defines a, b, c as $(-1)^a, (-1)^b, (-1)^c$. Show that using this quantum strategy, the players can win the game with certainty, i.e., $\omega_q(\text{GHZ}) = 1$.